Coefficient stability of d-matching polynomials

Vico Bonfioli*

July 2026

Abstract

Let $\mu_{d,G}$ be the d-matching polynomial of a finite graph G (the average matching polynomial over d-sheeted covers, Hall–Puder–Sawin). We show that for each fixed k the coefficient of $x^{|V|d-2k}$ in $\mu_{d,G}$ is a *polynomial in d of degree k*, and we compute its top two coefficients in closed form for every G:

$$[x^{|V|d-2k}]\,\mu_{d,G} = (-1)^k c_k(d), \qquad c_k(d) = \frac{|E|^k}{k!}\,d^k \;-\; \frac{|E|^{k-2}}{(k-2)!}\Big(\tfrac{|E|}{2} + P\Big)d^{k-1} + \cdots,$$

where $P = \sum_v \binom{\deg v}{2}$ is the number of 2-paths. We give the full c_1, c_2, c_3 as explicit polynomials in elementary graph invariants (edges, 2-paths, claws, triangles), and observe that at $k = 3$ the constant term counts triangles, $c_3(0) = -\operatorname{tri}(G)$; for $k \geq 4$ the constant term is a genuinely higher combinatorial invariant, not a simple count of k-edge cyclic subgraphs. The results are verified on the infinite cycle family (via the exact closed form $\mu_{d,C_n} = U_d(T_n(x/2))$) and on K_4 and $K_4 - e$. In contrast, an exhaustive search finds that K_4 admits *no* constant-coefficient recurrence in d of order ≤ 3 and x-degree ≤ 8 (unlike cycles), so the d-matching polynomial has no "cycle-like" global closed form of that shape; the coefficient stability above is the structure that survives.

1 Setup

For a finite graph $G = (V, E)$ and $d \geq 1$, a d-cover (or d-lift) H of G replaces each vertex by d copies and each edge by a perfect matching between the fibres given by a permutation in S_d. The *d-matching polynomial* $\mu_{d,G}$ is the average, over all $|E|$-tuples of independent uniform edge permutations, of the matching polynomial $\mu_H(x) = \sum_{k\geq 0}(-1)^k m_k(H)\,x^{|V(H)|-2k}$ [4], where $m_k(H)$ is the number of k-matchings of H [1]. Equivalently, by [1], $\mu_{d,G}$ is the expected characteristic polynomial of the "new" eigenvalues of a random d-cover. Since $|V(H)| = |V|d$, writing $c_k(d) := \mathbb{E}[m_k(H)]$ we have $\mu_{d,G} = \sum_k (-1)^k c_k(d)\,x^{|V|d-2k}$, and the object of study is the dependence of $c_k(d)$ on d.

For cycles, $\mu_{d,C_n} = U_d(T_n(x/2))$ with T, U the Chebyshev polynomials [2]; this gives exact data for all d. We also use exact data for K_4 (through $d = 5$) and $K_4 - e$ (the "diamond," through $d = 5$).

*Computations and an independent Lean/Mathlib formalization of related material: the `rama-notes` repository. Correspondence: `vicobonfioli@gmail.com`.

2 Coefficient stability

Theorem 1. *For each fixed $k \geq 0$, $c_k(d) = \mathbb{E}[m_k(H)]$ is a polynomial in d of degree k, with leading coefficient $|E|^k/k!$.*

Proof. Since $m_k(H) \leq \binom{|E|d}{k}$, the expectation $c_k(d) = \mathbb{E}[m_k(H)]$ is bounded above by a polynomial of degree k in d. That $c_k(d)$ is itself a polynomial in d is the specialization to k-matchings of the polynomiality of expected subgraph counts in random d-covers [3]: grouping the k-matchings of H by their projection to a k-edge sub-multigraph of G, each of the finitely many projection types contributes a polynomial in d. Proposition 2 evaluates the top two coefficients by inclusion–exclusion and in particular gives leading coefficient $|E|^k/k! \neq 0$; hence the degree is exactly k. The lower-order coefficients are polynomial in d by the same argument, but we do not reproduce them in closed form here (beyond c_1, c_2, c_3 in Proposition 3). □

Proposition 2 (top two coefficients)**.** *For $k \geq 2$, with $P = \sum_v \binom{\deg v}{2}$ the number of 2-paths of G,*

$$[d^k]\, c_k = \frac{|E|^k}{k!}, \qquad [d^{k-1}]\, c_k = -\frac{|E|^{k-2}}{(k-2)!}\Big(\frac{|E|}{2} + P\Big).$$

Proof. By inclusion–exclusion on the number of adjacent pairs, $m_k = \binom{M}{k} - p_2\binom{M-2}{k-2} + \cdots$, where $M = |E|d$ is the number of edges of H and $p_2 = P\,d$ is its number of 2-paths (both determined by the degree sequence of H, hence deterministic). The two displayed coefficients come from the top two terms: $\binom{M}{k}$ contributes $|E|^k d^k/k!$ and $-\binom{k}{2}|E|^{k-1}d^{k-1}/k! = -|E|^{k-1}d^{k-1}/(2(k-2)!)$; the term $-p_2\binom{M-2}{k-2}$ contributes $-P|E|^{k-2}d^{k-1}/(k-2)!$ at order d^{k-1}. Summing gives the claim. □

Proposition 3 (the first coefficients, explicitly)**.** *Let $S = \sum_v \binom{\deg v}{3}$ (number of claws $K_{1,3}$), $W = \sum_{uv \in E}(\deg u - 1)(\deg v - 1)$, and $t = \operatorname{tri}(G)$ the number of triangles. Then*

$$c_1 = |E|d, \quad c_2 = \binom{|E|d}{2} - P\,d, \quad c_3 = \binom{|E|d}{3} - P\,d(|E|d - 2) + (W + 2S)\,d - t.$$

Proof. $c_1 = m_1 = |E|d$ (edge count) and $c_2 = \binom{M}{2} - p_2$ are deterministic. For c_3, inclusion–exclusion gives $m_3 = \binom{M}{3} - p_2(M-2) + p_3 + 2(t_H + s)$, where $p_3 = Wd - 3t_H$ is the number of 3-edge paths, $s = Sd$ the number of claws, and t_H the number of triangles of H. All but t_H are deterministic; taking expectations with $\mathbb{E}[t_H] = t$ (each triangle of G lifts with expected multiplicity 1, being the expected number of fixed points of the uniform composition of its three edge-permutations) yields the stated cubic. □

Proposition 4 (the universal coefficients are exactly the top two)**.** *For $k \geq 2$ the two leading coefficients of $c_k(d)$ are given by the graph-independent formulas of Proposition 2 (functions of $|E|$ and P only). No further coefficient is universal in this sense: already $[d^1]c_3 = |E|/3 + 2P + W + 2S$ requires W, S, and $[d^2]c_4$ requires still more. Concretely, for the cycle C_n ($n \geq 5$) the exact fourth coefficient is*

$$c_4(C_n, d) = \tfrac{n^4}{24}d^4 - \tfrac{3n^3}{4}d^3 + \tfrac{107n^2}{24}d^2 - \tfrac{35n}{4}d,$$

whose d^2, d^1 coefficients $\frac{107}{24}n^2, -\frac{35}{4}n$ no formula in $(|E|, P)$ can reproduce; they are the cycle values of graph-specific combinations of subgraph counts (W, 3-paths, 4-cycles, claws, ...).

Thus c_1, c_2, c_3 are the fully explicit members; from c_4 on, only the top two coefficients have closed forms.

Remark 5 (topological constant term). The constant term $c_3(0) = -t$ (immediate from Proposition 3, verified on C_3–C_6, K_4, $K_4 - e$, the bowtie and the paw): the c_3 constant term counts triangles. Beyond $k = 3$ the constant term is no longer a simple invariant: direct computation gives $c_4(0) = 6, 1, 21, 20, 12, 8$ for C_3, C_4, the diamond, the bowtie, two disjoint triangles, and the paw, which depends on how triangles overlap and on pendant edges—so $c_4(0)$ is a genuine higher combinatorial invariant, not a function of $(\#C_4, \mathrm{tri})$ alone. The same holds for the middle coefficients: as the exact $c_4(C_n)$ above already shows, $[d^2]c_4$ is not a function of $(|E|, P)$, and expressing it requires several graph-specific subgraph counts (W, S, $\#C_4$, claws, $\ldots$). Thus, beyond the universal top two, the coefficients of c_k are delivered in principle by the projection/lift-count method but are genuinely multi-invariant (the quadratic subgraph basis), not low-order closed forms in $(|E|, P)$.

3 Verification and a contrast

Propositions 2–3 are verified for $k = 1, \ldots, 5$ on the cycles $C_3, \ldots, C_6$ (using the exact $\mu_{d,C_n} = U_d(T_n(x/2))$), on K_4 ($|E| = 6$, $P = 12$, $S = 4$, $W = 24$, $t = 4$), and on $K_4 - e$ ($|E| = 5$, $P = 8$, $S = 2$, $W = 12$, $t = 2$). The last two have $|E| \neq P$, so they independently exercise the P, S, W, t terms.

For K_4 we further observe, in contrast to cycles, that the μ_{d,K_4} satisfy *no* constant-coefficient linear recurrence in d with polynomial-in-x coefficients: solving $\mu_{d+2} = A(x)\mu_{d+1} + B(x)\mu_d$ from two instances yields A, B rational (not polynomial) in x, failing on the next instance, and an exhaustive search finds no order-≤ 3, degree-≤ 8 recurrence. Thus K_4 has no "cycle-like" closed form of the type $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$ proved for cycles in [2]; coefficient stability is the structure that persists in general. (This dichotomy is sharp: the top x-degree coefficients of μ_{d,K_4} are universal polynomials in d, while the constant term is $\mathbb{E}[\#\text{perfect matchings}]$, which we compute to grow like $(16/9)^d$.)

Formalization. The coefficient extraction that Proposition 2 rests on is machine-checked in Lean 4/Mathlib (`RamaLean/Paper4Coeff.lean`, standard axioms only, no `sorry`). Writing $L_E^{(k)} = \prod_{i<k}(E \cdot X - i)$, so that $\binom{Ed}{k} = L_E^{(k)}(d)/k!$, the file proves $\deg L_E^{(k)} \leq k$ (`L_natDegree_le`), the leading coefficient $[X^k]L_E^{(k)} = E^k$ (`L_coeff_top`), and the next one $[X^k]L_E^{(k+1)} = -\big(\sum_{i\leq k} i\big)E^k$ (`L_coeff_next`) — which is the $-\binom{k}{2}|E|^{k-1}$ of the proof. `binom_two_div_factorial` is the arithmetic $\binom{k}{2}/k! = 1/(2(k-2)!)$ that merges the two contributions, and `top_two_coeff` assembles them into the displayed formula. The inclusion–exclusion itself is machine-checked too, in `RamaLean/ConflictIE.lean`: for an arbitrary conflict relation on a finite type, `mCount_eq_ie` proves $m_k = \sum_S (-1)^{|S|}\#\{T : |T| = k,\ V(S) \subseteq T\}$ over the sets S of conflicting pairs, via the sign cancellation `freeIndicator_eq_alt_sum` and the exchange of summations `pairsIn_powerset_eq`. Nothing there uses bipartiteness or graphs, so the same statement serves the bipartite computation of the companion paper; taking conflict to be "shares an endpoint" gives the expansion used here. The identification of M with $|E|d$ and of p_2 with Pd is machine-checked in `RamaLean/CoverCounts.lean`. Neither needs the covering structure, only that the vertex map has all fibres of size d and preserves degrees:

`sum_over_fibers` shows any quantity accumulated over vertices as a function of the degree is multiplied by exactly d, and `edge_count` and `twoPath_count` are the cases $\varphi = \mathrm{id}$ and $\varphi = \binom{\cdot}{2}$. The only ingredient of Proposition 2 still outside Lean is then the degree bookkeeping that decides which inclusion–exclusion terms reach order d^{k-1}.

Reproducibility. Every numerical statement above is regenerated by a committed script in the `code/` directory of the `rama-notes` repository: `general_coefficient_formulas.py` (Propositions 2–3 on cycles, K_4, $K_4 - e$, the bowtie and the paw), `cycle_coefficient_stability.py` (the leading coefficient $|E|^k/k!$ against the exact closed form $\mu_{d,C_n} = U_d(T_n(x/2)))$, `k4_coefficient_stability.py` and `coeff_structure.py` (the coefficient polynomials themselves), `p4_exact.py`, and `k4_recurrence_search.py` (the exhaustive search establishing that K_4 admits no constant-coefficient recurrence of order ≤ 3 and x-degree ≤ 8).

References

[1] C. Hall, D. Puder, W. Sawin, *Ramanujan coverings of graphs*, Adv. Math. **323** (2018) 367–410.

[2] G. Cochran, C. Groothuis, A. Herring, R. Rohatgi, E. Stucky, *A new (combinatorial) proof of the commutativity of matching polynomials for cycles*, arXiv:1810.05889 (2018).

[3] N. Linial, D. Puder, *Word maps and spectra of random graph lifts*, Random Structures Algorithms **37** (2010), no. 1, 100–135.

[4] C. Godsil, I. Gutman, *On the matching polynomial of a graph*, in: Algebraic Methods in Graph Theory, 1981.